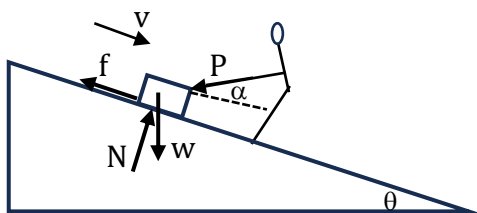


Solution: Group problem 3

Useful sketch:



Know:

w

μ_k

θ

α

$v = \text{const}$

Target: P

Physics approach: forces

$$\sum \vec{F} = m\vec{a}$$

Assume: air resistance is negligible

Because $v = \text{constant}$, $a = 0$.

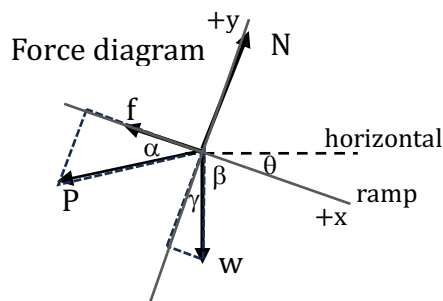
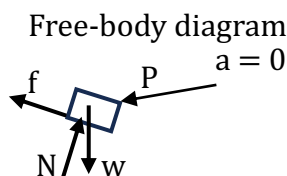
Frictional force: $f = \mu_k N$

Weight: $w = mg$

Choose crate as object of interest.

Choose coordinate system with x axis along the ramp.

Build a force diagram from a free-body diagram.



Don't know γ but know θ .

Determine γ from the force diagram by using the angles that add to 90° .

$\gamma + \beta = 90^\circ$ (the angle between the x and y axis).

Don't know β but know θ .

$\theta + \beta = 90^\circ$ (the angle between horizontal and vertical).

Comparing the two angle equations, $\gamma = \theta$.

Now the force equations are:

$$0 = w \sin \theta - f - P \cos \alpha \quad \checkmark \quad 0 = N - w \cos \theta - P \sin \alpha$$

2 equations with 3 unknowns (f , P , N).

However, f is related to N .

$$f = \mu_k N \quad \checkmark$$

The two force equations now have only 2 unknowns (P , N):

$$0 = w \sin \theta - \mu_k N - P \cos \alpha \quad 0 = N - w \cos \theta - P \sin \alpha$$

$$\sum F_x = 0$$

$$0 = w_x - f - P_x \quad \checkmark$$

$$\sum F_y = 0$$

$$0 = N - w_y - P_y \quad \checkmark$$

$$\sin \alpha = \frac{P_y}{P} \quad \checkmark \quad \cos \alpha = \frac{P_x}{P} \quad \checkmark$$

$$\sin \gamma = \frac{w_x}{w} \quad \cos \gamma = \frac{w_y}{w}$$

$$\sin \theta = \frac{w_x}{w} \quad \checkmark \quad \cos \theta = \frac{w_y}{w} \quad \checkmark$$

To solve for P, first eliminate the unknown N by solving for N in one force equation and then putting that solution in the other equation.

$$0 = N - w \cos \theta - P \sin \alpha$$

$$w \cos \theta + P \sin \alpha = N$$

$$0 = w \sin \theta - \mu_k N - P \cos \alpha$$

$$0 = w \sin \theta - \mu_k (w \cos \theta + P \sin \alpha) - P \cos \alpha$$

Isolate terms with P and put them on one side of the equation

$$0 = w \sin \theta - (\mu_k w \cos \theta + \mu_k P \sin \alpha) - P \cos \alpha$$

$$0 = w \sin \theta - \mu_k w \cos \theta - \mu_k P \sin \alpha - P \cos \alpha$$

$$\mu_k P \sin \alpha - P \cos \alpha = w \sin \theta - \mu_k w \cos \theta$$

Factor out P

$$P(\mu_k \sin \alpha - \cos \alpha) = w \sin \theta - \mu_k w \cos \theta$$

Divide both sides of the equation by $(\mu_k \sin \alpha - \cos \alpha)$

$$P = \frac{w \sin \theta - \mu_k w \cos \theta}{\mu_k \sin \alpha - \cos \alpha}$$

Check units:

$$[P] = \left[\frac{w \sin \theta - \mu_k w \cos \theta}{\mu_k \sin \alpha - \cos \alpha} \right] = [w] \left[\frac{\sin \theta - \mu_k \cos \theta}{\mu_k \sin \alpha - \cos \alpha} \right]$$

μ_k , sine, and cosine don't have units

$$[P] = [w]$$

Since P is a force and w is a force, they both have units of force: the units check.

Writing the answer in the simpler form by factoring out the weight,

$$P = w \frac{\sin \theta - \mu_k \cos \theta}{\mu_k \sin \alpha - \cos \alpha}$$

Is the behavior of the equation unreasonable?

If the weight of the crate increases, the force needed to keep it from accelerating increases. This is reasonable because it is more difficult to prevent a heavy crate from sliding down a ramp from speeding up than a lighter one.

Another check of the reasonableness of the answer is that if the push is along the ramp,

$$\alpha = 0$$

$$P = w \frac{\sin \theta - \mu_k \cos \theta}{\mu_k \sin \alpha - \cos \alpha} = \frac{\sin \theta - \mu_k \cos \theta}{0 - 1} = -w(\sin \theta - \mu_k \cos \theta)$$

I know if the ramp were level ground ($\theta = 0$), the force needed to keep the crate moving at a constant speed would be equal to the kinetic frictional force.

$$P = -w(0 - \mu_k 1) = \mu_k w$$

If the crate were moving on level ground, its weight would be equal to the force of the floor on it, the normal force (N).

$$P = \mu_k N$$

$\mu_k N$ is the frictional force.

$P = f$ as expected for pushing an object along a horizontal floor at a constant velocity.

Again, the solution gives a reasonable result for a situation where I know the answer.